

Full Sharded Data Parallel

Lorem Ipsum

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11. Sharding: split one big thing into pieces

One large model state

MODEL STATE

Story intuition

In games, shards are small pieces that together complete one item. FSDP uses the same idea for training state.

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Break it into shards

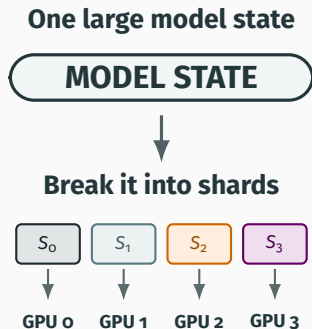


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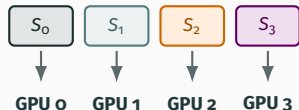
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MODEL STATE



Break it into shards



Story intuition

In games, shards are small pieces that together complete one item. FSDP uses the same idea for training state.

- Split one large state into pieces.
- Give different pieces to different GPUs.
- Reconstruct a full piece of the model only when computation needs it.

FSDP starts by **splitting** model state instead of **replicating** it.

12. What exactly is sharded?

DDP: replicated



Every GPU keeps a full copy.

FSDP: sharded

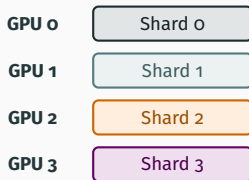
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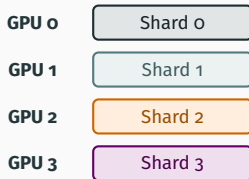
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P = parameters (weights), **G = gradients**, **O = optimizer states** (for Adam, moment estimates). FSDP can shard all three persistent model states.

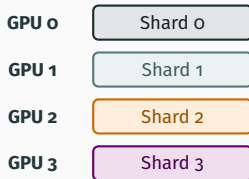
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Idealized persistent model-state memory per GPU scales roughly as $\frac{P+G+O}{N}$.

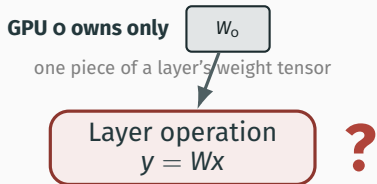
13. The first problem: a layer cannot compute from one shard

GPU 0 owns only



one piece of a layer's weight tensor

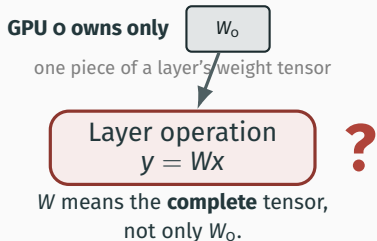
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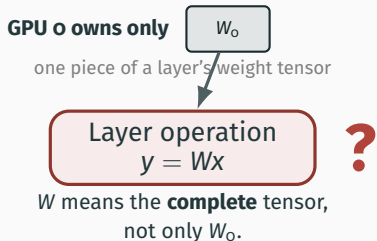
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$$W_0 + W_1 + W_2 + W_3$$

The equation shows four colored boxes representing weight shards: a grey box for W_0 , a light blue box for W_1 , an orange box for W_2 , and a purple box for W_3 . These boxes are separated by plus signs.

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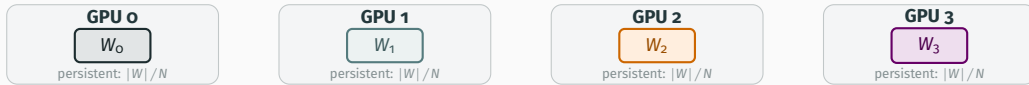
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So FSDP must **temporarily collect** the missing parameter shards before computation.

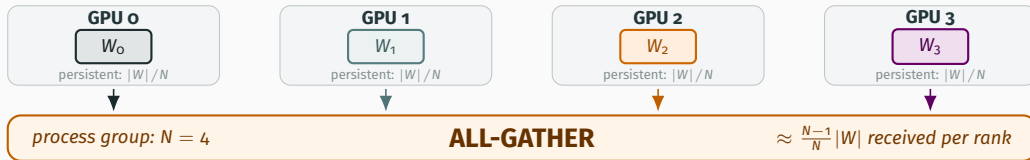
14. All-Gather: temporarily rebuild the current FSDP unit

(a) Before. Each rank persists one disjoint shard W_i of this unit's parameter tensor.



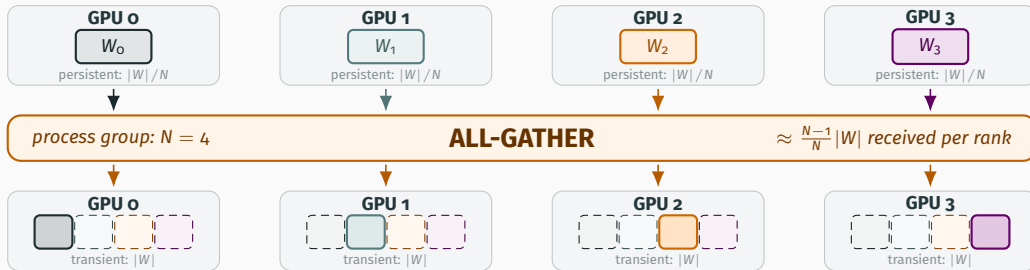
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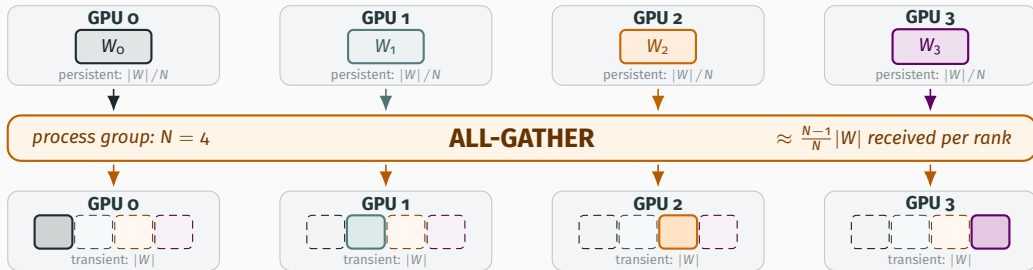
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(b) After. Every rank materializes $W = [W_0 \parallel W_1 \parallel W_2 \parallel W_3]$; solid = owned shard, dashed = gathered.

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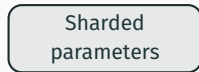


(b) **After.** Every rank materializes $W = [W_0 \parallel W_1 \parallel W_2 \parallel W_3]$; solid = owned shard, dashed = gathered.

All-Gather: **shards** → **full parameters**, but only for the unit currently being computed.

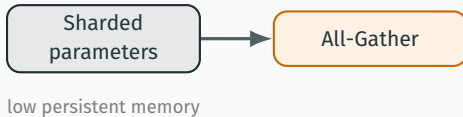
15. Forward pass: gather, compute, reshard

Sharded
parameters

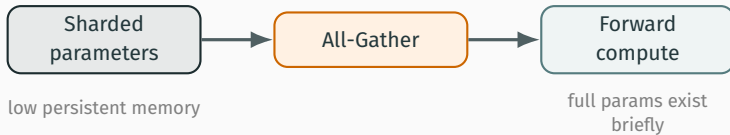


low persistent memory

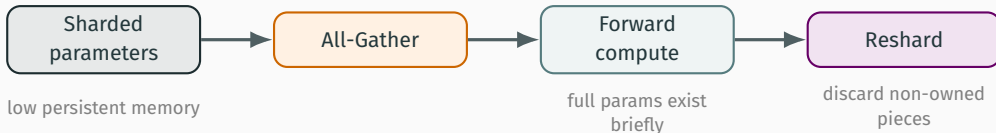
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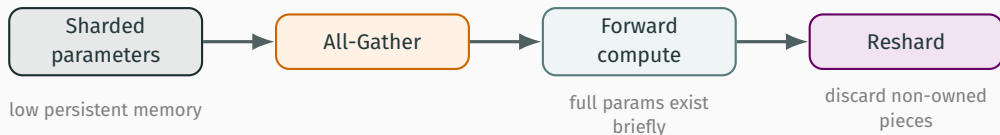
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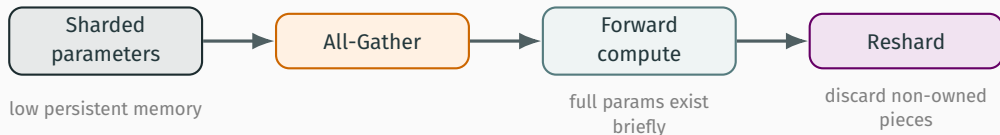
Resharding

After the forward computation, each GPU frees the full parameter and keeps only the shard it owns.

Why it saves memory

The expensive full representation is temporary; the sharded representation is the persistent one.

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Forward = **All-Gather** → **compute** → **reshard**.

16. Why gathering does not destroy the memory benefit

Layer-wise FSDP units

Transformer block 1

Transformer block 2

Transformer block 3

Transformer block 4

Other blocks can stay sharded.

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Block 1: full

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Important distinction

FSDP gathers only the **current FSDP unit** (for example, one Transformer block), not the entire model.

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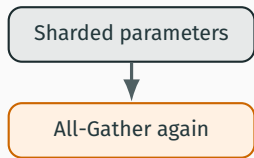
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FSDP controls peak memory by gathering **one unit at a time**.

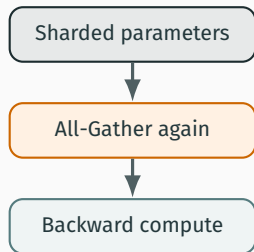
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Sharded parameters

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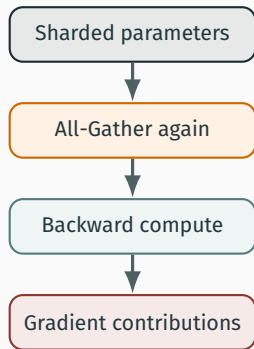
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Why gather again?

Backpropagation needs the current layer's parameters to compute gradients with respect to inputs and weights.

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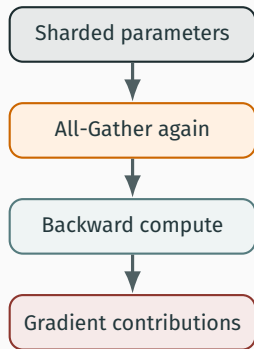
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Backward creates distributed gradient contributions; FSDP must combine them efficiently.

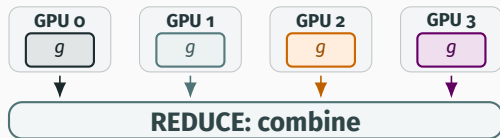
18. Reduce-Scatter: combine gradients, then split again

1. Gradient contributions



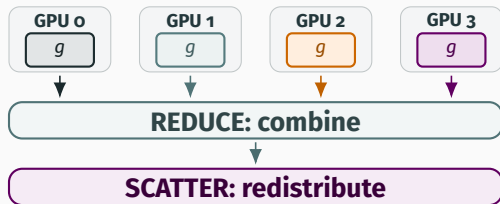
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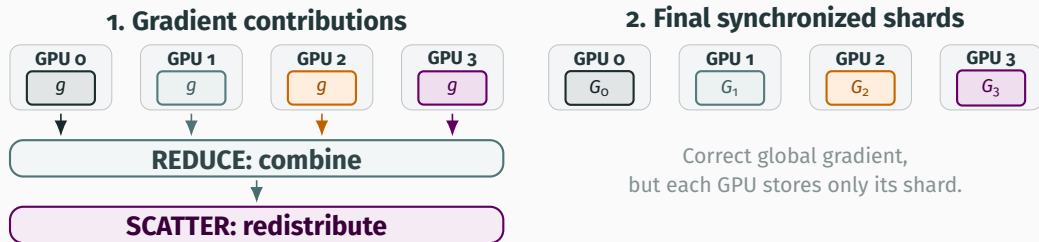


2. Final synchronized shards



Correct global gradient,
but each GPU stores only its shard.

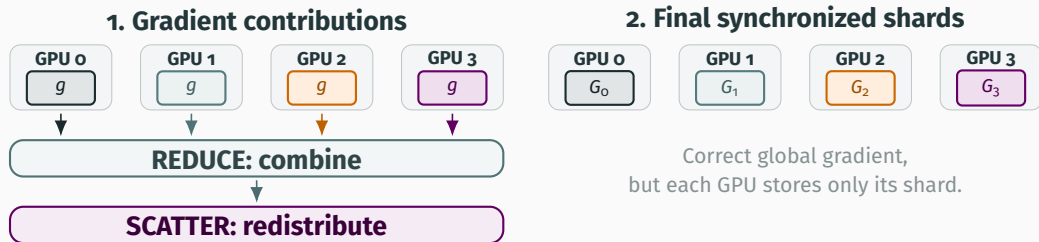
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Compare with DDP

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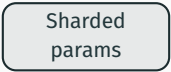


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Reduce-Scatter = **combine** gradient contributions, then **redistribute** only the owned shard.

19. The complete FSDP training cycle

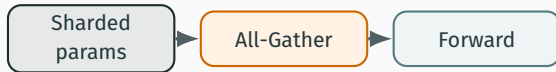


Sharded
params

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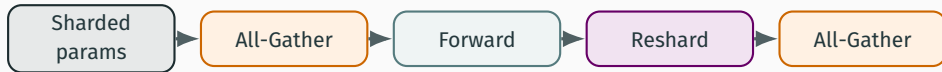
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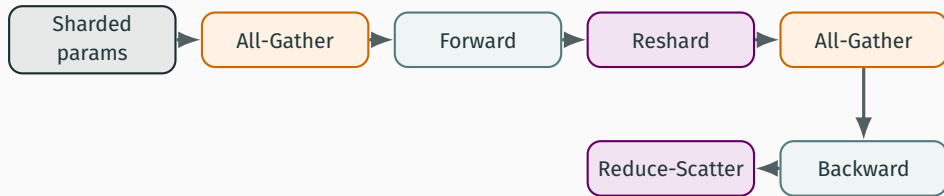
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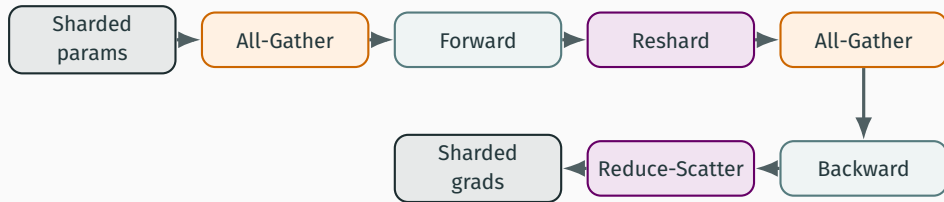
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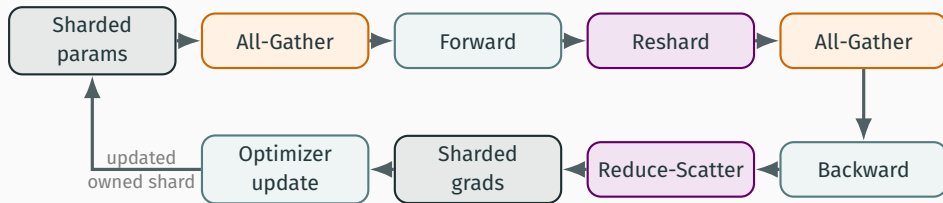
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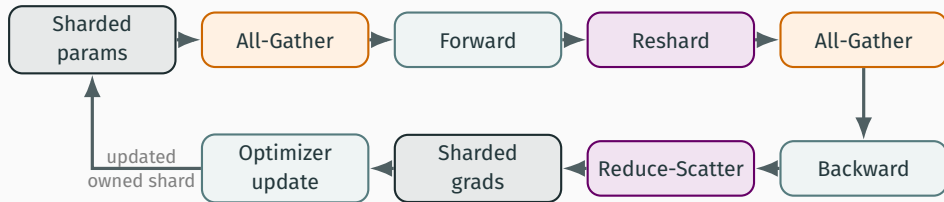
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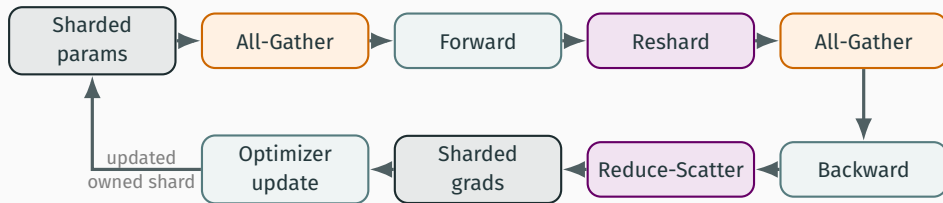


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Keep model state divided for memory; gather parameters briefly only when computation needs them.

20. Final trade-off: memory down, communication up

	DDP	FSDP
Model state	Replicated	Sharded
Memory/GPU	Higher	Lower
Parameters	Already local	All-Gather
Gradient sync	All-Reduce	Reduce-Scatter
Main pressure	Memory	Communication

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Overlap hides some cost

Compute block i

Gather block $i + 1$

Communication for the next unit can overlap with current computation.

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FSDP enables larger-model training by trading **memory** for **communication**.